
Hidden-Feature Matching and Left-Right Movement in Gait Representations

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A representation of an evolving system should support estimates of observable
2 movement as well as its training task. We examine this requirement in gait using
3 a small skeleton Joint-Embedding Predictive Architecture (JEPA). The model pre-
4 dicts hidden skeleton features; a separate regression estimates a signed contrast
5 between left and right movement speeds from its frozen encoder. In source-video
6 cross-validation on 625 clips from 93 videos, trained predictors more consistently
7 match hidden features from the correct clip than from another video. Yet with an
8 expanded feature summary, mean source-balanced R^2 for the movement estimate
9 falls from 0.22 at initialization to 0.10–0.11 after training. Motion-weighted and
10 connected-region masks do not establish an improvement over their own random
11 references. This focused negative finding connects latent prediction to evaluation
12 of observable movement: success on hidden-feature matching did not improve
13 the specified movement estimate under the tested training and readout procedure.
14 Future-only prediction remains a separate test.

15 1 Introduction

16 Walking is an evolving system: coupled body landmarks change position over time, and movement
17 depends on their trajectories. Gait rhythm and coordination motivate temporal modeling, although
18 video-derived coordinates describe only part of the physical process. Temporal world modeling aims
19 to represent such a system well enough to predict its observable behavior. A practical evaluation
20 question is whether a representation trained through prediction preserves information needed for a
21 specified movement measurement.

22 We study that question with S-JEPA [1], using a left–right contrast of median coordinate speeds.
23 Clinical gait asymmetry motivates the setting [2]; our contrast has no clinical validation. Our central
24 null is that JEPA training provides no improvement over the same encoder at initialization when a
25 frozen readout estimates this contrast on held-out source videos. The current expanded-readout and
26 mask comparisons are exploratory analyses on a repeatedly examined cohort; they do not retrospec-
27 tively become registered tests.

28 The contribution is a domain-specific evaluation of predictive representations, relevant to the rela-
29 bility of temporal foundation models. The present study establishes neither foundation-model scale
30 nor a gait simulator.

31 2 Study design

32 **Data and observable.** Our selected GAVD subset [3] contains 666 annotations, 642 matching
33 pose archives and, after quality checks, 625 clips from 93 source videos. Each archive has shape

34 $T_i \times 33 \times 4$: three estimated coordinates plus landmark visibility, with original frame numbers
 35 and frame rate. Coordinate units are image normalized with inferred depth; they are not calibrated
 36 physical speeds.

37 The movement path uses observed coordinates, pelvis centering and a body-width scale, without
 38 gap filling or resizing. For five pairs—shoulders, knees, ankles, heels and foot tips—we calculate
 39 median speeds $m_{L,k}, m_{R,k}$ using original timestamp differences. Both sides must be observed at
 40 both ends of at least eight common transitions for every pair. The signed target is

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

41 Anatomical reflection exchanges left/right landmarks and validity flags and flips the centered hor-
 42 izontal coordinate, giving $y(Mx) = -y(x)$. Hips define the pelvis and are excluded from the
 43 contrast because centering equalizes their speed magnitudes. The statistic does not measure phase,
 44 stride timing or loading; reversing a trajectory together with its sampling intervals preserves its speed
 45 magnitudes.

46 **Encoder input.** A landmark is observed when visibility is at least 0.45 and all coordinates are finite.
 47 A separate encoder path fills interior gaps of at most four samples, treating filled positions as valid.
 48 After pelvis centering and body-width normalization, coordinates and validity are linearly resized to
 49 64 positions by sequence index. Original time intervals are discarded. Resized validity must reach
 50 0.999; invalid coordinates are zero. Each joint’s four-position patch projects 12 coordinates to 96
 51 features, giving a 16×33 grid. A patch is valid only if all four positions are valid. Anatomical
 52 identities and prepared positions survive this operation; the original trajectory, duration and cadence
 53 need not.

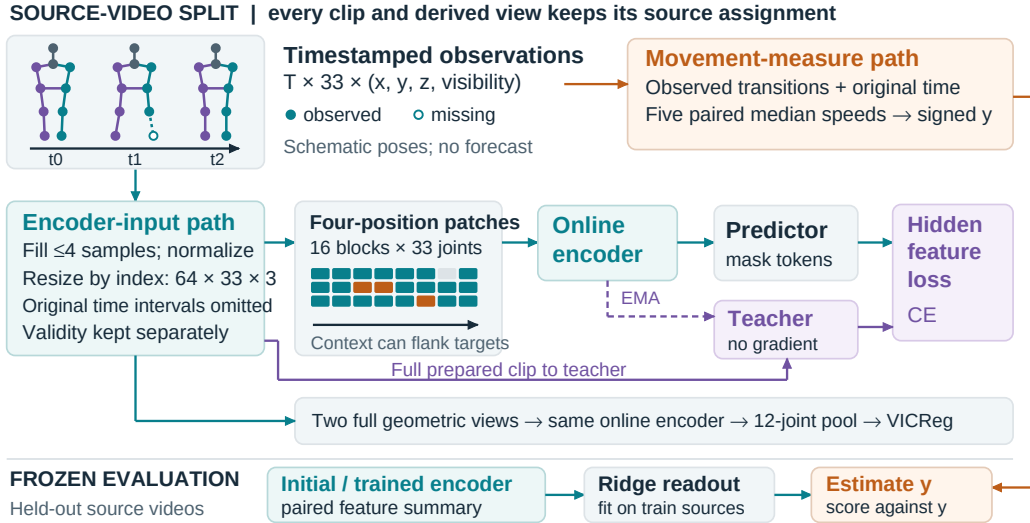


Figure 1: Two measurement paths and the completed JEPA training grid. Poses and mask cells are schematic. The movement target reaches only the separate regression evaluation. The online encoder has visible context on either side of a hidden position; the teacher and full-view regularizer access complete prepared clips.

54 **Training.** The grid uses 1,200 AdamW updates per encoder (architecture and settings in Appendix
 55 A). The online encoder zeroes hidden patch content before adding joint and position embeddings;
 56 the predictor substitutes learned mask tokens at target positions. Cross-entropy matches centered
 57 teacher feature distributions at hidden valid positions, averaged within clips and then across clips.
 58 A teacher follows the online weights by exponential moving average without gradients. A full-view
 59 VICReg term [4], weighted 0.05, pools twelve gait joints across two geometric views. Thus the total
 60 objective also accesses complete clips. Training uses CUDA BF16 and evaluation FP32.

We compare three motion conditions and two region conditions across five folds and five seeds: 125 encoders. The motion conditions use random sampling, a MAMP-style sampler [5], and a robust motion mixture. Their target count is half the batch’s smallest valid twelve-joint gait pool, rounded down, sampled across all 33 joints. Connected regions hide six connected landmarks over eight of sixteen blocks, with a separate count-matched random reference. Each comparison matches initial weights, source draws, views and update counts. No reflection augmentation or reflection loss appears in this grid.

Held-out readout. Five outer folds keep each video’s clips and derived views together. Fresh encoders fit only outer-training videos with source-balanced sampling. All overlapping windows, mirrors and augmentations inherit the source split. The encoder is frozen while ridge regression, a linear model with a coefficient penalty, estimates y . Three inner source folds select the penalty using only outer-training labels; imputation and scaling are fitted within each inner training set and then refitted with the regression on all outer-training sources. Inner-validation clips may have contributed without labels to outer-training encoder fitting.

Both encoders use bilateral mean features (960 inputs), expanded with standard deviations, absolute adjacent-block feature changes and ten validity-support fractions (2,890 inputs). We pool outer-test predictions within each seed, weighting each video equally, then average the five R^2 values. Paired 95% source-bootstrap intervals resample videos with their clips and seed predictions together (2,000 resamples). They condition on the fitted models and existing splits. Repeated seeds and mask draws do not add independent sources.

3 Findings

Movement readout falls after training. The expanded initial representation reaches mean $R^2 = 0.223$, compared with 0.101–0.114 for trained teachers (Figure 2). All five conditions decrease in every seed, with greater mean absolute error. Paired trained-minus-initial intervals are below zero in every condition. Final online encoders also score below their matched initial encoders.

Training mask	Mean R^2	ΔR^2	95% source interval
Random (motion)	0.114	-0.109	[-0.170, -0.041]
MAMP-style	0.113	-0.110	[-0.165, -0.051]
Motion mixture	0.114	-0.108	[-0.171, -0.042]
Random (region)	0.109	-0.113	[-0.166, -0.060]
Connected region	0.101	-0.122	[-0.183, -0.055]

Table 1: trained teacher versus matched initialization; exploratory paired intervals conditional on fitted models. Differences precede rounding.

The compact initial readout scores 0.071; expanding it increases R^2 by 0.152. The added components mix variation, feature changes and observation support, so the gain cannot be assigned specifically to temporal order. The initial control retains the pretrained pose detector, input preparation, anatomical identities and supervised regression. A 264-value direct-pose summary reaches $R^2 = 0.035$; this limited baseline does not bound stronger coordinate readouts.

Correct-clip features become easier to match. Each diagnostic compares source-weighted mean squared feature error against the correct clip and valid targets from another source, holding the evaluated clips and hidden positions fixed. All 375 trained checks favor correct targets; 33 of 75 initial checks do so. The trained checks cover 125 fitted models under three evaluation masks (left-leg gap, right-leg gap and scattered gap). Initial checks cover 25 matched fold/seed controls under the same three masks, with copies shared between experiment families counted once. This preference could involve pose, viewpoint or missingness as well as motion. Teachers and target distributions change during training, so raw training losses cannot rank representation quality across conditions.

Changing mask geometry does not establish better readout. The inspection draws confirm that motion samplers select higher-motion targets and that connected regions remove immediate temporal brackets. Motion masks hide approximately 17% of valid all-landmark tokens; regions hide 9.9%. Their readout comparisons therefore use distinct random references. MAMP-style, motion-mixture and region differences are respectively -0.0008 , $+0.0005$ and -0.009 , with in-

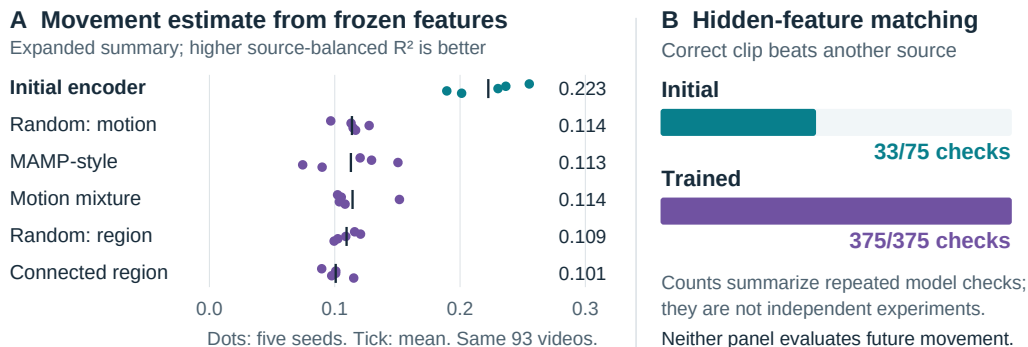


Figure 2: Different outcomes from the same fitted models. A: frozen movement readout; points are five seed scores, and short vertical marks their mean. B: correct-clip feature error is lower in 375 of 375 trained diagnostic checks, versus 33 of 75 distinct initial checks. These checks reuse models and videos.

107 tervals $[-0.020, 0.015]$, $[-0.015, 0.015]$ and $[-0.033, 0.018]$. Intervals spanning zero leave benefits
 108 uncertain, rather than establishing equivalence.

109 **Preparing input changes the measurement.** A saved comparison recalculates the target after input
 110 preparation: 623 clips from 92 sources have finite values on both paths, with source-weighted sign
 111 agreement 70.4% and calculation-agreement $R^2 = 0.218$. This is agreement between two calcula-
 112 tions of the observable, not a learned model score. Neither the achievable readout accuracy nor an
 113 individual preprocessing cause follows from it.

114 4 Interpretation and next experiment

115 The two evaluations answer different questions. JEPA training improves preference for the correct
 116 clip’s hidden features, while a fixed regression procedure estimates its left–right speed contrast less
 117 accurately. Poorer readout performance does not imply complete destruction of movement information.
 118 In particular, 49/125 teacher fits and 96/125 online fits select the largest ridge penalty, 10,000.
 119 Wider training-only selection, support-only controls and summaries of matched dimension would
 120 help separate feature geometry from readout limitations. Person identities, reuploads, pose errors
 121 and recording cues remain possible explanations beyond what source holdout controls.

122 A next experiment would decode future coordinates from past-only inputs at common valid end-
 123 points and physical-time horizons. Fit and select decoders on training sources only; compare last-
 124 position, constant-velocity, direct-pose, initial-encoder and mismatched-training-future controls. Fu-
 125 ture coordinates and visibility must not change prefix preparation, normalization, masks or predic-
 126 tions. The retained past-feature readout and future-feature decoding studies have synthetic training
 127 evidence; the real-data horizon census establishes eligibility only. Decoding observed future-teacher
 128 features diagnoses the decoder; decoding features predicted from the past evaluates forecasts. Mul-
 129 tiple horizons from one prefix are not recursive rollouts.

130 The finding is conditional on this cohort, recipe and readout. Forecasting and simulation require
 131 tests of future behavior beyond the whole-clip contrast.

132 5 Related work

133 Skeleton action recognition motivates learned feature targets [1], motion targets [5] and connected-
 134 body masking [6]. We vary masks within one JEPA objective without reproducing these complete
 135 methods. Video anticipation and planning [7], together with transformation-sensitive representa-
 136 tion learning [8], show why evaluation depends on the intended behavior. Here the endpoint is an
 137 observed gait quantity: comparing its recovery before and after training complements those task
 138 evaluations.

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172 A Measurement and implementation details

173 The completed recipe uses 96 features, encoder/predictor depths 4/2, four attention heads, batch size
174 20, 1,200 AdamW updates, learning rate 0.001, weight decay 0.05 and teacher momentum 0.999.

175 The archive coordinates are $x = (x_0 + uW_c)/W$, $y = (y_0 + vH_c)/H$, and $z = dW_c/W$, where crop
176 origin and size are (x_0, y_0) , W_c, H_c , frame size is W, H , and u, v, d are pose estimates. Unequal
177 image-axis scaling and inferred depth remain after body normalization. The scale is the median of
178 valid shoulder and hip widths. The movement path requires observed hips; the encoder uses the
179 median valid pelvis midpoint for missing hips (zero if none is available). No boundary extrapolation
180 fills gaps.

181 The teacher receives the unaugmented prepared clip; the masked online path receives a geometric
182 view. Two full online views use rotations up to 8 degrees in the horizontal-depth plane and transla-
183 tions up to 0.03 in image-plane coordinates. Target/predictor temperatures are 0.06/0.10; the teacher
184 center uses momentum 0.9. Gradient norm is clipped at 1. The twelve-joint regularizer pool is
185 shoulders, hips, knees, ankles, heels and foot tips. Its projector is a separate training component.
186 It combines invariance, variance and covariance penalties with internal weights 25, 25 and 1, then
187 overall weight 0.05.

188 The expanded summary uses common left/right validity within each pair; increments require con-
189 secutive common valid blocks. Each pair contributes differences and sums of means, standard de-
190 viations and mean absolute feature increments (six 96-vectors), plus two support fractions. Thus
191 $5 \times 6 \times 96 + 10 = 2,890$. Missing support gives zero summary values. Temporal positions in
192 this summary are prepared block indices, not original timestamps. Ridge penalties are selected from
193 0.01, 0.1, 1, 10, 100, 1000, 10000.

194 Source-balanced R^2 uses weights $w_{vi} = 1/(Vn_v)$ and weighted test mean $\bar{y}_w$:

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

195 The test mean defines this metric’s denominator, not a deployable fitted baseline. Outer test folds
196 contain 19, 19, 19, 18 and 18 videos and 189, 182, 72, 77 and 105 clips. Source IDs and split
197 membership are checked; no persistent participant identities or complete cross-upload content com-
198 parison establish independence across different IDs. Annotation retrieval/version records are incom-
199 plete.

200 B Supporting results and reproducibility

201 These exploratory percentile intervals use bootstrap seed 812, retain all five fitted seeds and have no
202 familywise or development-selection adjustment. The numerical supplement preserves aggregate
203 seed scores and recorded intervals. The executed report identifies 125,000 prediction rows and 200
204 pooled score rows; a saved audit records their recomputation. Raw predictions and checkpoints
205 are unavailable in the revision environment, so this revision checks aggregate consistency without
206 claiming to repeat that bootstrap or inference.

207 A separate reflection-augmentation comparison has real-data summary evidence: augmentation re-
208 duces the specified teacher-token reflection discrepancy by $\Delta q = -0.008 [-0.010, -0.007]$, while
209 its readout difference is $\Delta R^2 = 0.004 [-0.006, 0.013]$. This uses its own recipe, four inner readout
210 folds and compact summary; it is not pooled with the motion grid. An explicit reflection loss has
211 synthetic evidence only. Likewise, a readout forced to take the form $g(x) = [h(x) - h(Mx)]/2$
212 reverses sign algebraically without establishing learned movement information.

213 GAVD annotations describe the cohort, not affected side or treatment response. Uncalibrated coor-
214 dinates and absent clinical reference measurements preclude a clinical interpretation of y . Public
215 annotation access does not authorize redistribution of separately hosted videos or derived poses;
216 neither is included in the supplement.

217 **AI assistance.** An AI assistant supported author-side evidence reconciliation, literature checking,
218 manuscript revision, figure programming and adversarial checks. No new model training was per-

219 formed for this revision. Responsibility for the submitted text, figures, claims and references remains
220 with the authors.